

Q

Cross Correlation

$$x(n) = \{1, 1, 2, 2\}$$

$$y(n) = \{1, 3, 1\}$$

$$r_{xy} = ?$$

$$y(-n) = \{1, 3, 1\}$$

	1	3	1
1	1	3	1
1	1	3	1
2	2	6	2
2	2	6	2

lowest = -2

high = 3

$$r_{xy} = \{1, 4, 6, 9, 8, 2\}$$

Auto Correlation

lowest = -3

highest = 3

$$x(n) = \{1, 2, 3, 4\}$$

$$x(-n) = \{4, 3, 2, 1\}$$

	4	3	2	1
1	4	3	2	1
2	8	6	4	2
3	12	9	6	3
4	16	12	8	4

$$r = \{4, 11, 20, 30, 20, 8, 4\}$$

$$y(n) = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k - \sum_{k=0}^{n-1} \left(\frac{1}{2}\right)^k = \sum_{k=n}^{\infty} \left(\frac{1}{2}\right)^k$$

$$\sum_{k=n}^{\infty} \left(\frac{1}{2}\right)^k = \left(\frac{1}{2}\right)^n \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \left(\frac{1}{2}\right)^n \cdot 2 = \left(\frac{1}{2}\right)^{n-1}$$

$$\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k u(k-n) =$$

$$y(n) = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k u(k) u(n-k)$$

$$x(n) = u(n) \quad y(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$z_n = \left[\frac{1 - \left(\frac{1}{4}\right)^{n+1}}{1 - \left(\frac{1}{4}\right)} \right]$$

$$z_n = \sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^k$$

$$z_n = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k u(n-k)$$

$$z_{n-k} = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k u(n-k-k) = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k u(n-2k)$$

2.41 a) $y(n) = \sum_{k=0}^{\infty} h(k) x(n-k)$

$$x(n) = 2^n u(n) \quad h(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$c) x_3[n] = \begin{cases} \left(\frac{1}{2}\right)^n, & n=0,2,4,\dots \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} X_3(\omega) &= \sum_{n=-\infty}^{\infty} x_3[n] e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} x_3[n] e^{-j\omega n} = \sum_{n=0,2,4,\dots}^{\infty} \left(\frac{1}{2}\right)^n e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{2m} e^{-j\omega 2m} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n e^{-j2\omega n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{4} e^{-j2\omega}\right)^n \\ &\Rightarrow \\ &= \frac{1}{1 - \frac{1}{4} e^{-j2\omega}} \end{aligned}$$

$$(2) \quad x[n] = \left(\frac{1}{4}\right)^{|n|}$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

Stability Condition of an LTI Discrete-Time System

- Example - Consider a causal LTI discrete-time system with an impulse response

$$h[n] = (\alpha)^n \mu[n]$$

- For this system

$$S = \sum_{n=-\infty}^{\infty} |\alpha^n| \mu[n] = \sum_{n=0}^{\infty} |\alpha|^n = \frac{1}{1-|\alpha|} \quad \text{if } |\alpha| < 1$$

- Therefore $S < \infty$ if $|\alpha| < 1$ for which the system is BIBO stable
- If $|\alpha| = 1$, the system is not BIBO stable

Causality Condition of an LTI Discrete-Time System

- Example - The factor-of-2 interpolator defined by

$$y[n] = x_u[n] + \frac{1}{2}(x_u[n-1] + x_u[n+1])$$

is **noncausal** as it has a noncausal impulse response given by

$$\{h[n]\} = \{0.5 \quad \underset{\uparrow}{1} \quad 0.5\}$$

(5)

causal system

$$b) y[n] = \sum_{k=n}^{\infty} x[n-k] \quad \begin{matrix} k=1 \\ n=1 \end{matrix}$$

$$= x[n-k] + x[(n-1)-k] + x[(n-2)-k] + \dots$$

$$= x[0] + x[1] + x[2] + \dots$$

~~Non~~ causal

Stable or unstable:

$$Q1- y[n] = \frac{1}{1-x[n-2]}$$

Taking assumption:

$$\text{let, } |x[n-2]| \leq M_x < \infty$$

$$y[n] = \left| \frac{1}{1-M_x} \right|$$

Mx

$$\text{If } M_x = 1$$

$$y[n] = \infty$$

So at $M_x = 1$ given signal is unstable
Hence unstable.

$$Q2- y[n] = A x[n]$$

$$\text{let, } |x[n]| \leq M_x < \infty$$

$$x[l] = a \sum_{k=-\infty}^{\infty} x[k] x[k - (l - D)] + \sum_{n=-\infty}^{\infty} v[n] x[n - l]$$

Thus,

$$x_{xy} = x_{xx}[l - D] + x_{vx}[l]$$

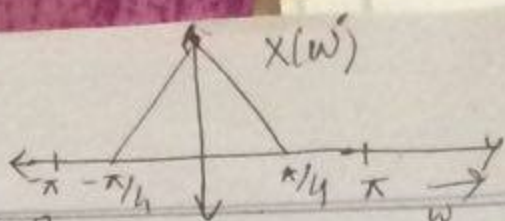
Hence we should observe peak at

$$l - D = 0$$

$$l = D$$

4.23

$$a) y_1[n] \triangleq \begin{cases} x[n], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$$



$$c) y_3[n] \triangleq \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$$

Find $Y_1(w)$, $Y_3(w)$ in terms of $X(w)$

$$X(w) = \sum_{n \text{ even}} x[n] e^{-jwn} = \left[\sum_{n=-\infty}^{\infty} \left[\frac{1+(-1)^n}{2} \right] x[n] e^{-jwn} \right]$$

$$\begin{aligned} Y_1(w) &= \sum y_1[n] e^{-jwn} = \sum_{n \text{ even}} y_1[n] e^{-jwn} + \sum_{n \text{ odd}} y_1[n] e^{-jwn} \\ &= \frac{1}{2} [X(w) + X(w - \pi)] \end{aligned}$$

$$\begin{aligned} x[n] &\rightarrow (D) \rightarrow y[n] \\ y[n] &= x[2n] \\ Y(w) &= \frac{1}{D} \sum_{k=0}^{D-1} X\left(\frac{w-2\pi k}{D}\right) \end{aligned}$$

$$b) y_2[n] = x[2n]$$

$$Y_2(w) = \frac{1}{2} \left[X\left(\frac{w}{2}\right) + X\left(\frac{w-2\pi}{2}\right) \right]$$

* H.W 4

$x(t)$ has Nyquist Rate = $\omega_0 = 2B$
 $x(t) \longleftrightarrow X(\omega)$

Q2- a) $y(t) = x(t) + x(t-1) \xrightarrow{F} Y(\omega)$

$$Y(\omega) = X(\omega) + e^{-j\omega} X(\omega) = X(\omega)(1 + e^{-j\omega})$$

$$F_s = \omega_0 = 2B$$

b) $y(t) = \frac{dx}{dt} \longleftrightarrow Y(\omega)$

$$Y(\omega) = j\omega X(\omega)$$

$$F_s = \omega_0 = 2B$$

c) $y(t) = x^2(t) \longleftrightarrow Y(\omega)$
 $Y(\omega) = \frac{1}{2\pi} [X(\omega) * X(\omega)]$

$$B_{\text{new}} = 2B$$

$$F_s = 2\omega_0 = 4B$$

d) $y(t) = x(t) \cos(\omega_0 t) \longleftrightarrow Y(\omega)$

$$2\omega_0$$

$$Y(\omega) = \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$$

$$F_s = 2 \left[\overset{\omega_0}{\cancel{B}} + \omega_0 \right] = 2 \left[\overset{\omega_0}{\cancel{B}} + \overset{\omega_0}{\cancel{B}} \right] = 2 \left[\overset{\omega_0}{\cancel{B}} + \overset{\omega_0}{\cancel{B}} \right] = 2 \left[\overset{\omega_0}{\cancel{B}} + \overset{\omega_0}{\cancel{B}} \right]$$

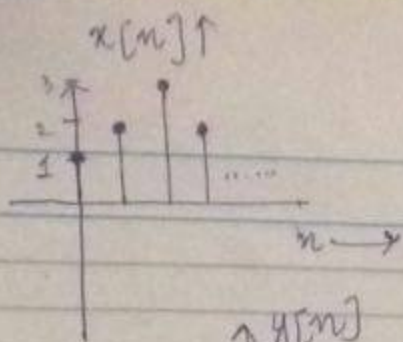
e) $y(t) = x(at) \longleftrightarrow Y(\omega)$

$$Y(\omega) = \frac{1}{a} X\left(\frac{\omega}{a}\right)$$

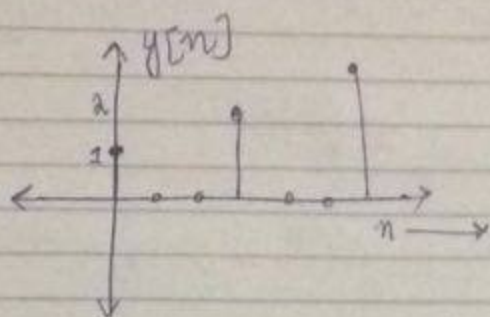
$$x(at) \longleftrightarrow \frac{1}{a} X\left(\frac{\omega}{a}\right)$$

$$B = \frac{\omega_0}{a} = \frac{2B}{a}$$

$$F_s = 4B$$



$L=3$



$$Y(\omega) = ?$$

$$= \sum_{n=-\infty}^{\infty} y[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} y[Ln] e^{-j\omega Ln}$$

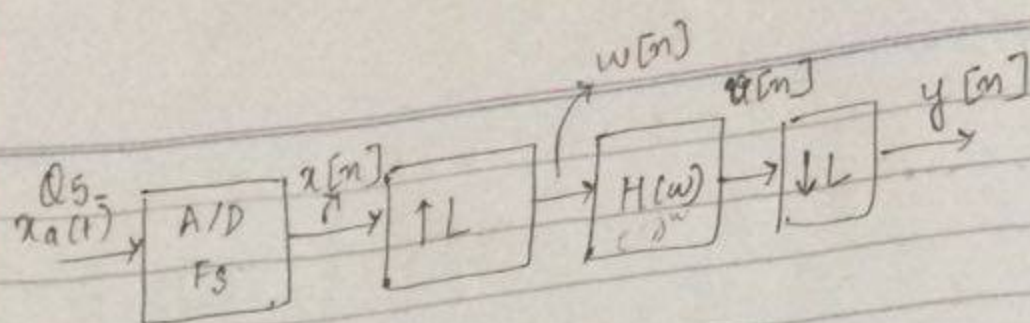
$$= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega Ln} = X(L\omega)$$

Q7- $x^*[-n]$ is to be expressed in form of $X(\omega)$

DTFT ($x^*[-n]$)

$$= \sum_{n=-\infty}^{\infty} x^*[-n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x^*[n] e^{j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} (x[n] e^{-j\omega n})^* = \left(\sum_{n=-\infty}^{\infty} x[n] e^{j\omega n} \right)^*$$



$$x_a(f) = 0 \text{ for } |f| > F_s/2$$

$$H(\omega) = \begin{cases} e^{-j\omega} & |\omega| \leq \pi/L \\ 0 & \pi/L < |\omega| \leq \pi \end{cases}$$

$$F_s = \frac{1}{T_s}$$

$$x[n] = x_a(nT_s)$$

$$w[n] = x_a\left(\frac{n}{LF_s}\right)$$

Now, as we know

$$x[n-n_0] \longleftrightarrow e^{-j\omega n_0} X(\omega) \quad \text{--- (1)}$$

~~using above Eq (1)~~ using above Eq (1)

$$v[n] = w[n-1]$$

$$= x_a\left(\frac{n-1}{LF_s}\right) = x_a\left(\frac{n}{LF_s} - \frac{1}{LF_s}\right)$$

Now,

$$y[n] = v[nL]$$

$$= x_a\left(\frac{nL}{LF_s} - \frac{1}{LF_s}\right)$$

$$= x_a\left(nT_s - \frac{T_s}{L}\right)$$

(4)

$$y[n] = A M_x < \infty \quad \forall M_x$$

$$y[n] = M_y < \infty$$

System BIBO stable

$$a) y[n] = x^*[n]$$

$$\text{Let, } x[n] \leq M_x < \infty$$

$$y[n] = M_x^2 < \infty \quad \forall M_x; M_x \in \mathbb{R}$$

$$y[n] = M_y$$

System BIBO stable

$$b) y[n] = \frac{e^{x[n]}}{x[n-1]}$$

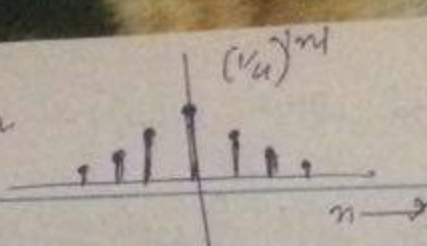
$$x[n] \leq M_x < \infty$$

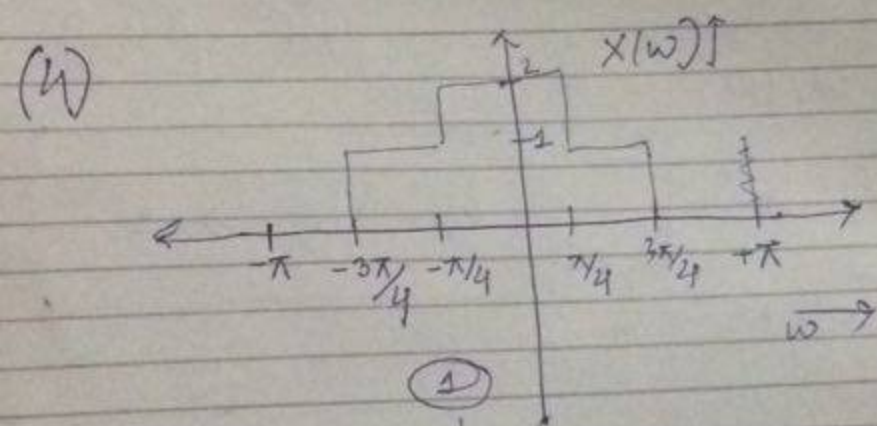
$$x[n-1] = 0$$

$$x[n] = M_x$$

$$y[n] = \frac{e^{M_x}}{0} = \infty$$

Unstable system

$$\begin{aligned}
 &= \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n e^{-j\omega n} + \sum_{n=-\infty}^{-1} \left(\frac{1}{4}\right)^n e^{-j\omega n} \\
 &= \frac{1}{1 - \frac{1}{4}e^{-j\omega}} + \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n e^{j\omega n} \\
 &= \frac{1}{1 - \frac{1}{4}e^{-j\omega}} + \frac{\frac{1}{4}e^{j\omega}}{1 - \frac{1}{4}e^{j\omega}}
 \end{aligned}$$




$x[n] = ?$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

DTFT: $\frac{\sin(\frac{\omega}{2}n)}{\pi n} \leftarrow \text{DTFT} \times \text{rect}\left(\frac{\omega}{\pi}\right)$

$$X(\omega) = X_1(\omega) + X_2(\omega) = \text{rect}\left(\frac{\omega}{6\pi/4}\right) + \text{rect}\left(\frac{\omega}{2\pi/4}\right)$$

IDTFT $\{X(\omega)\}$

Q 2.65: $x_a(t)$ = Transmitted signal

(a) $y_a(t)$ = Received signal = $a x_a(t - t_d) + \underbrace{v_a(t)}_{\text{Noise}}$

a) Delay measurement computing cross correlation.

$r_{xy}(l)$ -

$$x(n) = x_a(nT)$$

$$y(n) = y_a(nT) = a x_a(nT - DT) + v_a(nT)$$

~~Sample T as very small~~

$$y(n) = a x_a(nT - D) + v_a(nT)$$

$$\triangleq a x[n - D] + v[n]$$

Thus in received signal there is a delay of D samples because of round trip time delay.

Moreover,

$$DT = \frac{2R}{c}$$

c = speed of light

R = Range to target

$\Rightarrow$ Use cross correlation to estimate D then R can be estimated as

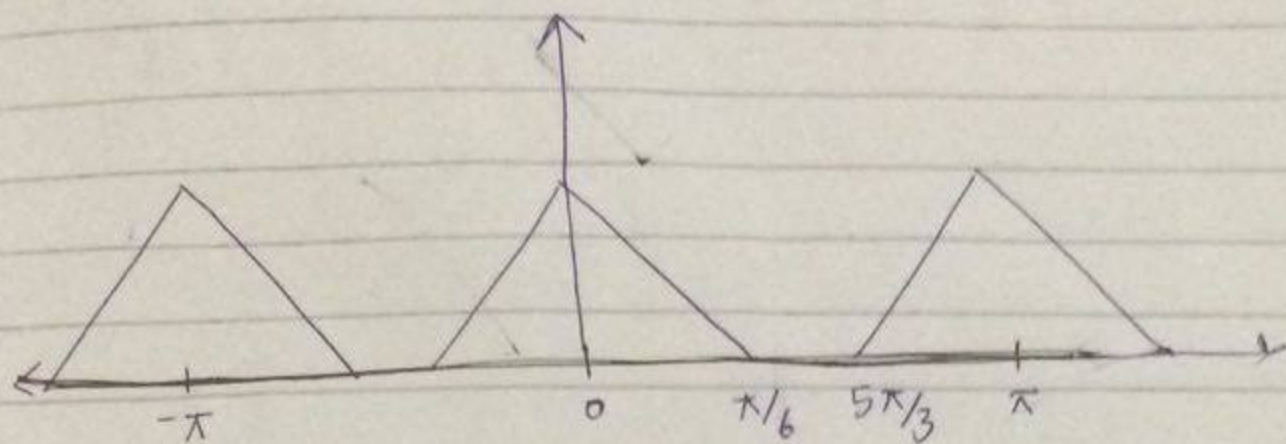
$$R = \frac{cDT}{2}$$

$$r_{xy}[l] = \sum_{n=-\infty}^{\infty} y[n] x[n-l] = \sum_{n=-\infty}^{\infty} (a x[n-D] + v[n]) x[n-l]$$
$$+ \sum_{n=-\infty}^{\infty} v_a[n] x[n-l]$$

lets do a change of variables.

$$k = n - D \Rightarrow \boxed{n = k + D}$$

$X_3(\omega)$:



$$y(n) = \sum_{m=0}^3 x(m)h(m-n)$$

(10)

$$n = -2: \quad y(-2) = x(0)h(+2) + x(1)h(1+2) \\ + x(2)h(2+2) + x(3)h(3+2)$$

$$= (8 \times 6) + 0 + 0 + 0$$

$$= 48$$

$$n = -1:$$

$$y(-1) = x(0)h(1) + x(1)h(2) + 0$$

$$= (8 \times 3) + (9 \times 6)$$

$$= 78$$

$$y(0) = 71$$

$$y(1) = 60$$

$$y(2) = 17$$

$$y(3) = 12$$

$$y(n) = \{48, 78, 71, 60, 17, 12\}$$

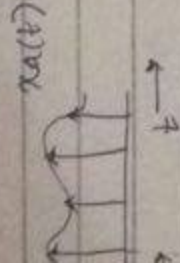
Discrete Time Fourier Transform: (DTFT)

$$e^{j\omega n} = e^{j(\omega + 2\pi)n}$$

$$X(\omega) = \sum_n x[n] e^{j\omega n}$$

main sampling:

$$x_s(t) = x_a(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$



$$a) x_1[n] = \left(\frac{1}{2}\right)^n u[n+3]$$

$$X_1(\omega) = \sum_{n=-\infty}^{\infty} x_1[n] e^{-j\omega n} = \sum_{n=-3}^{\infty} \left(\frac{1}{2}\right)^n e^{-j\omega n}$$

$$= \sum_{n=-3}^{\infty} \left(\frac{1}{2} e^{-j\omega}\right)^n = \left(\frac{1}{2} e^{-j\omega}\right)^{-3} \sum_{n=0}^{\infty} \left(\frac{1}{2} e^{-j\omega}\right)^n$$

$$= \left(\frac{1}{2} e^{-j\omega}\right)^{-3} \sum_{n=0}^{\infty} \left(\frac{1}{2} e^{-j\omega}\right)^n$$

$$\Rightarrow \left| \frac{1}{2} e^{-j\omega} \right| < 1$$

$$= \left(\frac{1}{2} e^{-j\omega}\right)^{-3} \frac{1}{1 - \frac{1}{2} e^{-j\omega}}$$

$$c) y[n] = \cos(x[n])$$

$$x[n] \leq M_x < \infty$$

$$y[n] = \cos(M_x) < \infty \quad \forall M_x$$

Hence stable

$$Q \quad y[n] = \sum_{k=0}^n x[k] \quad \neq (n+1)M_x$$

$$\Rightarrow \text{let, } x[n] \leq M_x < \infty$$

$$y[n] = (n+1)M_x$$

$$-\infty < n < \infty$$

$$y[n] = \infty$$

Hence unstable

$$Q: y[n] = nx[n]$$

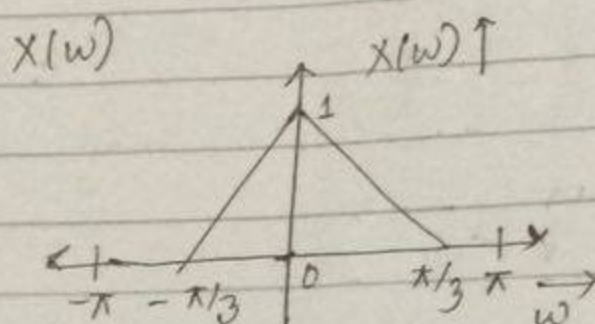
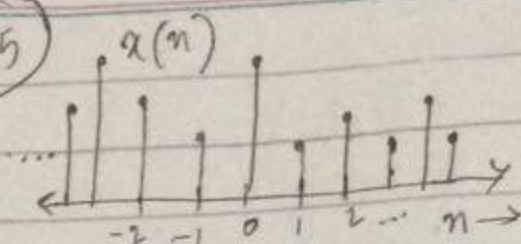
$$x[n] \leq M_x < \infty$$

$$-\infty < n < \infty$$

$$y[n] = nM_x = \infty \quad \text{at } n = \infty$$

Hence unstable

11.5



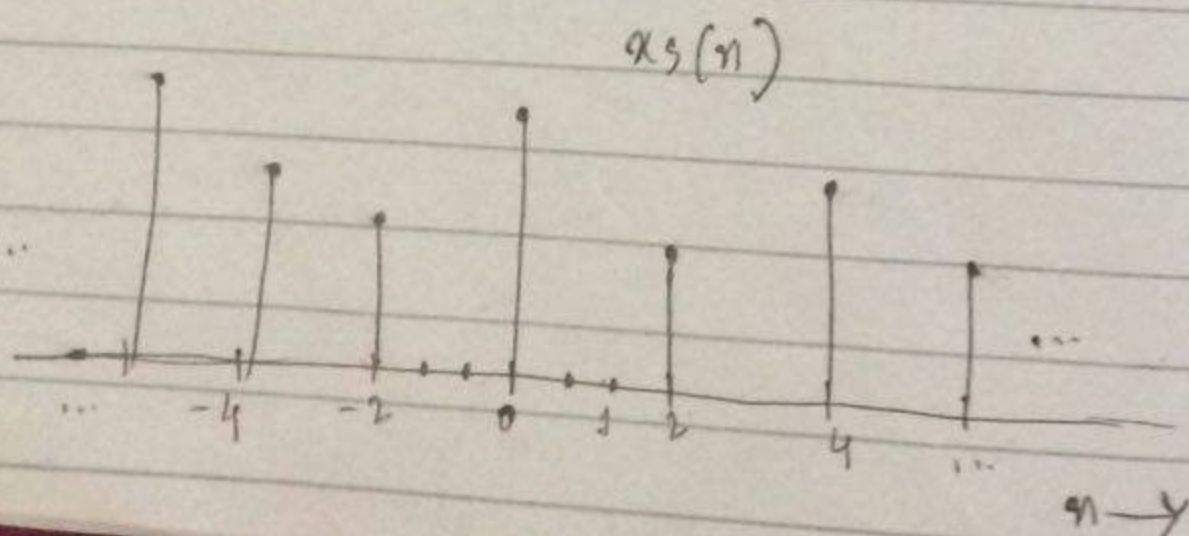
a) ^{up} Sampling $x(n)$ with $D=2$

$$x_s(n) = \begin{cases} x(n), & n = 0, \pm 2, \pm 4, \dots \\ 0, & n = \pm 1, \pm 3, \pm 5, \dots \end{cases}$$

compute & sketch $x_s(n)$, $X_s(w)$

$$x_s(n) = x\left[\frac{n}{2}\right] \quad n = 0, \pm 2, \pm 4, \dots$$

$$X_s(w) = X(2w)$$



$$u(n) = x_1(n) - x_1(n-10)$$

$$x(n) = u(n) - u(n-10) = y_1(n) - y_1(n-10)$$

$$\Rightarrow h(n) = a^n u(n)$$

$$y_1(n) = \sum_{k=-\infty}^{\infty} x(n-k) h(k)$$

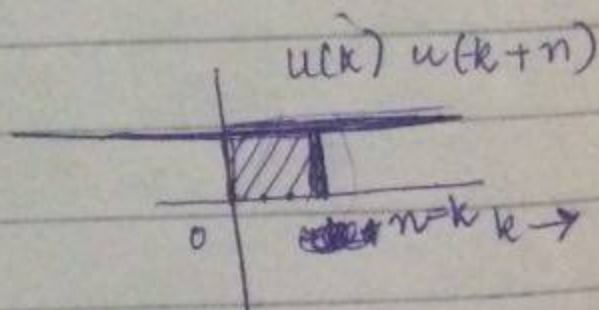
$$= \sum_{k=-\infty}^{\infty} u(n-k) a^k u(k)$$

Prob
2.39

$$= \sum_{k=-\infty}^{\infty} a^k u(k) u(n-k)$$

$$= \sum_{k=-\infty}^{\infty} a^k u(k) u(k+n)$$

$$= \sum_{k=0}^{n+1} a^k$$



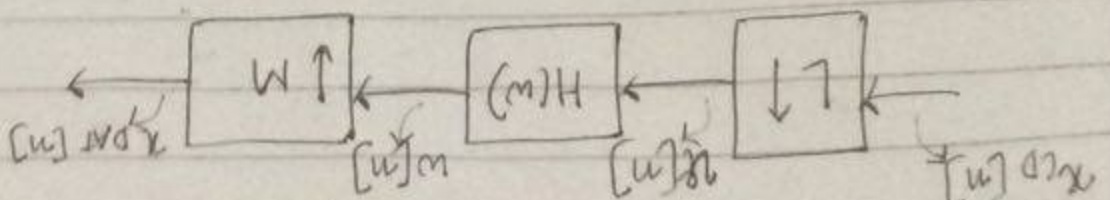
$$= \frac{1 - a^{n+1}}{1 - a}$$

$$= \frac{1}{1-a} \left[(1 - a^{n+1}) - (1 - a^{n-9}) \right]$$

Q3: $F_s = 48 \text{ KHz} \rightarrow x_{\text{dat}}[n]$

we need $44.1 \text{ KHz} \rightarrow x_{\text{cd}}[n]$

Find L, M and filter $H(w)$ for this conversion?



$48000 = 2^7 \cdot 3 \cdot 5^3$ and $44100 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7$

$$\frac{L}{M} = \frac{2^7 \cdot 3 \cdot 5^3}{2^2 \cdot 3^2 \cdot 5^2 \cdot 7} = \frac{147}{160}$$

If we upsample by 160 and downsample by $M=147$ we achieve desired result

LPF required $\nRightarrow w_c = \min\left(\frac{\pi}{L}, \frac{\pi}{M}\right) = \frac{\pi}{160}$

Filter gain should be $L=160$

2.3.4 Impulse Response = ?

$$h_1 = \left(\frac{1}{2}\right)^n u(n)$$

$$h_2 = \left(\frac{1}{4}\right)^n u(n)$$

$$h(n) = h_1(n) * h_2(n)$$

$$= \sum_{k=-\infty}^{\infty} h_1(k) h_2(n-k)$$

$$= \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k}$$

$$= \left(\frac{1}{4}\right)^n \sum_{k=0}^{\infty} \left(\frac{4}{2}\right)^k$$

$$= \left(\frac{1}{4}\right)^n \sum_{k=0}^n (2)^k$$

$$= \left(\frac{1}{4}\right)^n \left[\frac{1 - 2^{n+1}}{1 - 2} \right]$$

$$= \left(\frac{1}{4}\right)^n (2^{n+1} - 1)$$

$$= \text{IDTFT} \{X_1(\omega)\} + \text{IDTFT} \{X_2(\omega)\}$$

$$= \text{IDTFT} \left\{ \text{rect} \left(\frac{\omega}{6\pi/n} \right) \right\} + \text{IDTFT} \left\{ \text{rect} \left(\frac{\omega}{2\pi/n} \right) \right\}$$

$$x[n] = \frac{\sin \left(\frac{3\pi n}{4} \right)}{\pi n} + \frac{\sin \left(\frac{\pi}{4} n \right)}{\pi n}$$

(could have used
Eq (1) for
last step also)

⑤ $X(\omega) = \cos^2(\omega)$

$$= \left[\frac{1}{2} e^{j\omega} + \frac{1}{2} e^{-j\omega} \right]^2$$

$$= \frac{1}{4} e^{j2\omega} + \frac{1}{4} e^{j2\omega} + \frac{1}{2}$$

$$\boxed{\delta[n-n_0] \xleftrightarrow{\text{DTFT}} e^{j\omega n_0}}$$

$$= \frac{1}{4} \delta[n+2] + \frac{1}{4} \delta[n-2] + \frac{1}{2} \delta[n]$$

⑥ $x[n] \xleftrightarrow{\text{DTFT}} X(\omega)$

$$y[n] = \begin{cases} x[n/2], & n = 0, \pm 1, \pm 2 \\ 0, & \text{otherwise} \end{cases}$$

He No y
Play
Bachay

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$$N_1 = 4$$

$$N_2 = 4$$

$$d) x_4(n) = \{1, 2, 3, 4\} \quad h_4(n) = \{1, 2, 3, 4\}$$

$$\begin{array}{cccccc} & 1 & 2 & 3 & 4 & 0 \\ 4 & 3 & 2 & 1 & & & & & & \\ & 4 & 3 & 2 & 1 & & & & & \\ & & 4 & 3 & 2 & 1 & & & & \\ & & & 4 & 3 & 2 & 1 & & & \\ & & & & 4 & 3 & 2 & 1 & & \\ & & & & & 4 & 3 & 2 & 1 & \\ & & & & & & 4 & 3 & 2 & 1 \\ & & & & & & & 4 & 3 & 2 & 1 \end{array}$$

$$y(0) = 1$$

$$y(1) = 4$$

$$y(2) = 10$$

$$y(3) = 20$$

$$y(4) = 25$$

$$y(5) = 24$$

$$y(6) = 16$$

$$y(7) = 0$$

$$y(n) = \{1, 4, 10, 20, 25, 24, 16\}$$

Ranges for $n_i(m) = ?$

$$m_L = x_L = 0 \quad m_H = x_H = 3$$

$$0 \leq m \leq 3$$

$$n_L = x_L - (n_L + N_1 - 1) = 0 - 0 - 4 + 1 = -3$$

$$n_H = n_L + N_1 + N_2 - 2 = -3 + 4 + 4 - 2 = 3$$

$$-3 \leq n \leq 3$$

$$n(-3) = 1$$

$$n(-2) = 4$$

$$n(-1) = 10$$

$$n(0) = 20$$

$$n(1) = 25$$

$$n(2) = 24$$

$$n(3) = 16$$

Some kind
again

$$n_i(n) = \{1, 4, 10, 20, 25, 24, 16\}$$

$$n_i(n) = \{1, 4, 10, 20, 25, 24, 16\}$$

Stability Condition of an LTI Discrete-Time System

- **BIBO Stability Condition** - A discrete-time system is BIBO stable if and only if the output sequence $\{y[n]\}$ remains bounded for all bounded input sequence $\{x[n]\}$
- An LTI discrete-time system is BIBO stable if and only if its impulse response sequence $\{h[n]\}$ is absolutely summable, i.e.

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

$$x[n] = ?$$

$$X(\omega) = 2 \operatorname{rect}\left(\frac{\omega}{2\pi}\right) - \operatorname{rect}\left(\frac{\omega}{14\pi/8}\right) - \operatorname{rect}\left(\frac{\omega}{12\pi/8}\right) \\ + \operatorname{rect}\left(\frac{\omega}{6\pi/8}\right) + \operatorname{rect}\left(\frac{\omega}{2\pi/8}\right)$$

$$x[n] = \frac{2 \sin(\pi n)}{\pi n} - \frac{\sin\left(\frac{7}{8}\pi n\right)}{\pi n} -$$

$$\frac{\sin\left(\frac{3}{4}\pi n\right)}{\pi n} + \frac{\sin\left(\frac{3}{8}\pi n\right)}{\pi n} + \frac{\sin\left(\frac{1}{8}\pi n\right)}{\pi n}$$

$$x[n] = \dots$$

$$[N-1]x \dots x [1-N]$$

$$[N-u]x \dots x [n-u]x \dots x [n-1]x \dots x [n-N]$$

$$c) \quad y[n] = \prod_{k=1}^N x[n-k] \quad n=1$$

Non-causal

$$y[-2] = x[-2] - x[4+2]$$

$$n=2$$

$$y[1] = x[1] - x[0]$$

$$n=1$$

$$d) \quad y[n] = x[n] - x[n^2-n]$$

causal

$$y[-1] = x[-1] + x[-4] + x[-11]$$

$$n=-1$$

$$y[1] = x[1] + x[-2] + x[-9]$$

$$n=1$$

$$e) \quad y[n] = x[n] + x[n-3] + x[n-10]$$

Non-causal

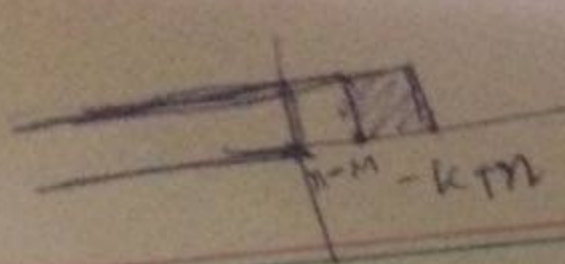
$$y[-1] = x[-1]$$

$$n=-1$$

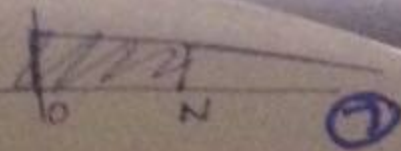
$$y[1] = x[1]$$

$$n=1$$

$$f) \quad y[n] = x[n]$$



-99x
a



$$h_1(n) = a^n [u(n) - u(n-N)]$$

$$h_2(n) = [u(n) - u(n-M)]$$

Impulse Response = ?

$$h(n) = a^n [u(n) - u(n-N)] * [u(n) - u(n-M)]$$

$$= \sum_{k=-\infty}^{\infty} a^k [u(k) - u(k-N)] \cdot [u(n-k) - u(n-k-M)]$$

$$= \sum_{k=0}^N a^k \cdot u(k+n) - \cancel{a^{k+n}} u(-k+(n-M))$$

$$\sum_{k=n-M}^N a^k = \frac{a^{n-M} - a^{N+1}}{1-a}$$

$$y[n] = x[2n+1]$$

$$Y(\omega) = \sum_{n=-\infty}^{\infty} y[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x[2n+1] e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \frac{1}{2} [1 - (-1)^n] x[n] e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} \frac{1}{2} x[n] e^{-j\omega n}$$

$$= \frac{1}{2} \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \frac{1}{2} X(\omega) - \frac{1}{2} X(\omega - \pi)$$

H.W. 4

20/10/15



when

⑥

d) $y[n] = \sum_{k=-\infty}^n x[k]$

let, $x[n] \leq Mx < \infty$

$x[n] = u[n]$

$y[n] = n+1$

$-\infty < n < \infty$

at $n = \infty$

$y[n] = \infty$

⇒ Hence unstable system

e) $y[n] = \log(1+x[n])$

let, $x[n] \leq Mx < \infty$

$y[n] = \log(1+Mx) < \infty$

$y[n] = My < \infty$

⇒ Hence stable

b) $y[n] = x[n] * \cos(n\pi/8)$

correlation

$$x(n) = \{ \overset{\downarrow}{8}, 9, 2, 3 \}$$

$$N_1 = 4$$

$$h(n) = \{ \overset{\downarrow}{4}, 3, 6 \}$$

$$N_2 = 3$$

~~q(1) = 2.1~~

$$y(n) = x(n) \oplus h(n)$$

$\downarrow$

$$N = N_1 + N_2 - 1 = 6$$

$$y(n) = \sum_{m=-\infty}^{\infty} x(m)h(m-n)$$

$$\Rightarrow n_L = x_L = 0 \quad n_h = x_h = 3$$

$$0 \leq n \leq 3$$

$$\Rightarrow n_L = x_L - (n_L + N_2 - 1)$$

$$= 0 - (0 + 3 - 1) = -2$$

$$\Rightarrow n_h = n_L + N_1 + N_2 - 2$$

$$= -2 + 4 - 2 = -4$$

$$-2 < n < +3$$

$$[-2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3]$$

Indexes

$$= 6$$

$$m_L = x_L$$

$$m_h = x_h$$

15

$$r(n) = \sum_{m=-\infty}^{\infty} x(m)h(m-n)$$

$$x(n) = \{ \overset{\downarrow}{1}, 2, 4 \} \quad h(n) = \{ \overset{\downarrow}{1}, 1, 1, 1, 1 \}$$

Finding Range of m & n

$$N_1 = 3$$

$$N_2 = 5$$

$$N = N_1 + N_2 - 1 = 3 + 5 - 1 = 7$$

$$\Rightarrow m_L = x_L = 0$$

$$m_h = x_h = 2$$

$$0 \leq m \leq 2$$

$$\Rightarrow n_L = x_L - (h_L + N_2 - 1)$$

$$= 0 - (0 + 5 - 1) = -4$$

$$n_h = n_L + N_1 + N_2 - 2$$

$$= -4 + 3 + 5 - 2 = 2$$

$$-4 \leq n \leq 2$$

$$y(-4) = 1$$

$$y(-3) = 3$$

$$y(-2) = 7$$

$$y(-1) = 7$$

$$y(0) = 7$$

$$y(1) = 6$$

$$y(2) = 4$$

using answers from convolution

$$r(n) = \{ 1, 3, 7, 7, \overset{\downarrow}{7}, 6, 4 \}$$

some kharud
baki

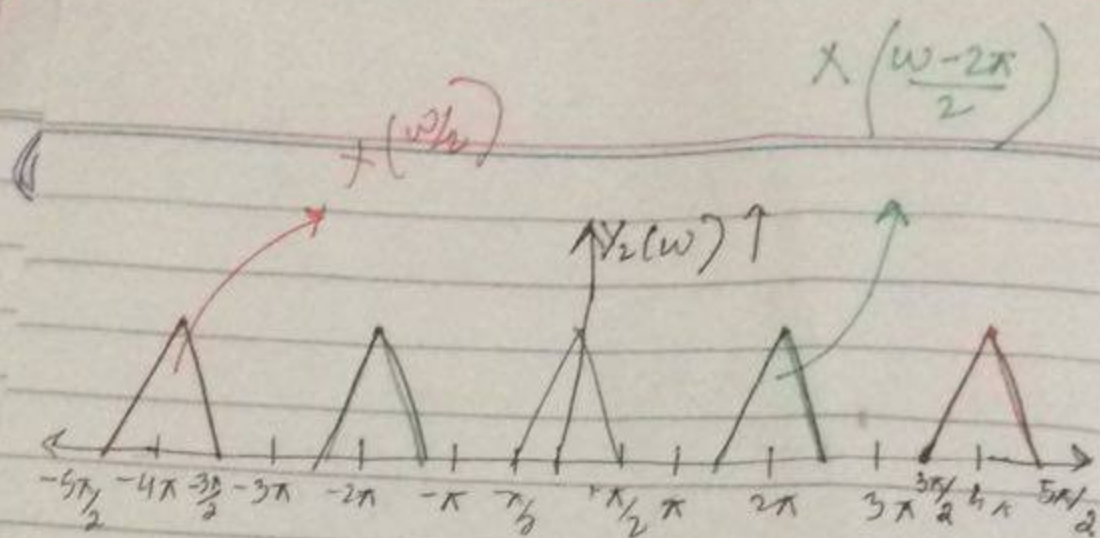
2.54 i) $y_i(n) = ?$ ii) correlation $r_i(n) = ?$

a) $x_1(n) = \{1, 2, 4\}$ $h_1(n) = \{1, 1, 1, 1, 1\}$

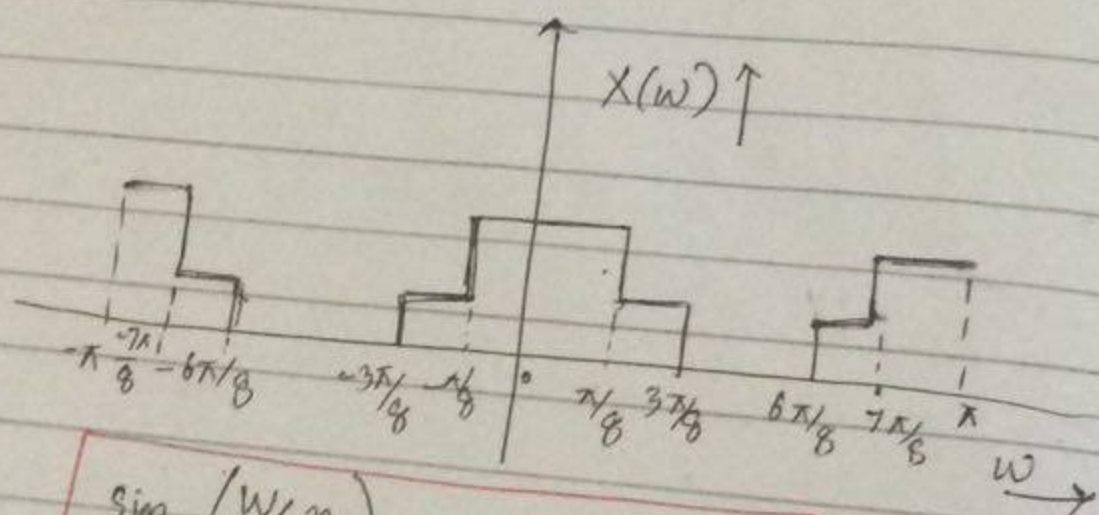
[illegible]

$$y(n) = \begin{cases} 0, & n < 0 \\ 1, & n = 0 \\ 3, & n = 1 \\ 7, & 2 \leq n \leq 4 \\ 6, & n = 5 \\ 4, & n = 6 \\ 0, & n > 6 \end{cases}$$

$\Rightarrow$ ~~is it a sequence?~~ $y(n) = \{1, 3, 7, 7, 7, 6, 4\}$
 $x(n) = ?$



4.2



$$\frac{\sin\left(\frac{\omega}{a}n\right)}{\pi n}$$

DTFT

$$\longleftrightarrow \text{rect}\left(\frac{\omega}{W}\right)$$

$\Rightarrow 9\pi$ Radian hai

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